

Unit 10, Lesson 2: Interpreting Key Features of Absolute Value Functions



Benchmark

MA.912.AR.4.3 Given a table, equation, or written description of an absolute value function, graph that function and determine its key features.



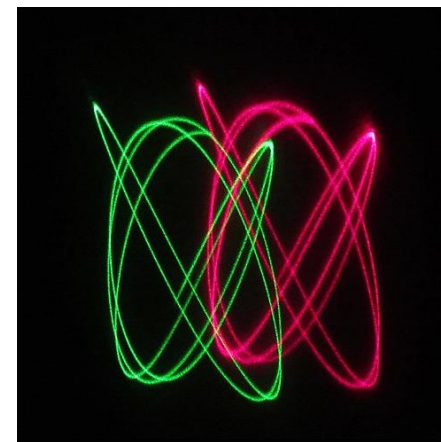
Learning Targets

- ☐ I can graph an absolute value function given a written description.
- ☐ I can determine key features of an absolute value function.



10.2.1 Warm-Up: Notice and Wonder

1. Cymatics is the study of wave phenomena, sound, and their visual representations. The picture shows a visual representation of sound created using two different lasers. The pattern that is formed is known as the Lissajous pattern, which is formed by movement in two directions at different rates.



- a. What do you notice?
- b. What do you wonder?

2. French physicist Jules Antoine Lissajous used sounds of different frequencies to vibrate a mirror. A beam of light reflected on the mirror would trace patterns on the mirror. Lissajous found that the pattern depends on the frequency.
Circle the graph that you think shows the sound with higher frequency.





10.2.2 Collaboration: Graphing Real-World Applications

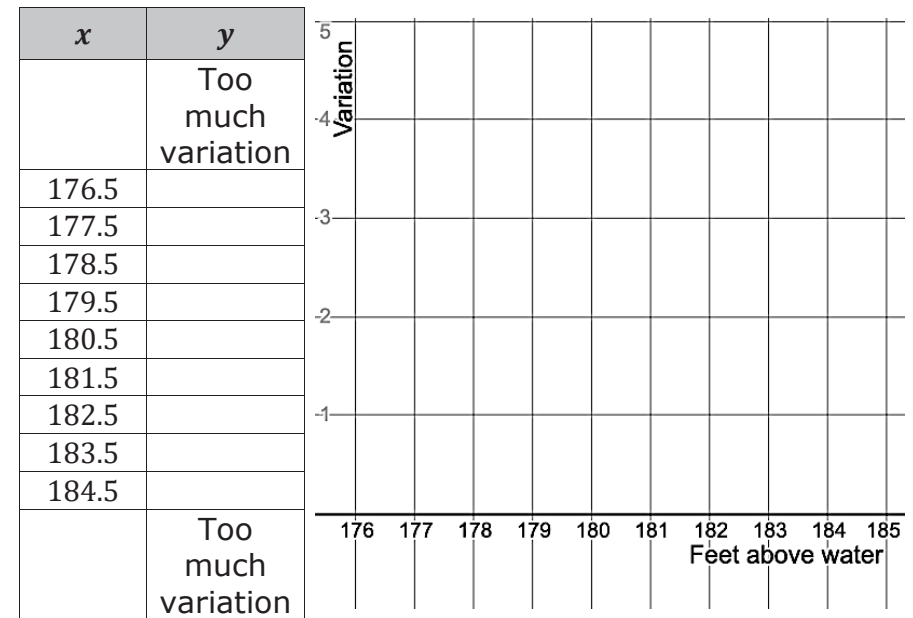
The clearance below the Sunshine Skyway Bridge over Tampa Bay, Florida, is 180.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 4 ft above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and y is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $y = |x - 180.5|$.



1. Discuss with your partner the maximum and the minimum clearance between the bridge and the water.

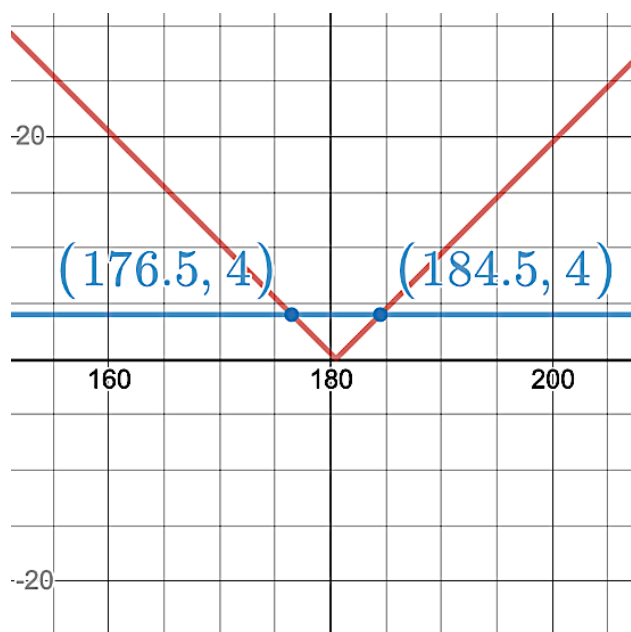
Minimum Clearance (ft)	Maximum Clearance (ft)

2. Complete the table and graph the absolute value equation to find the variation from the normal length.



3. Interpret the vertex of $y = |x - 180.5|$.

4. The equations $y = |x - 180.5|$ and $y = 4$ are shown.



Work with your partner to explain how the graph shows the minimum and maximum clearance.



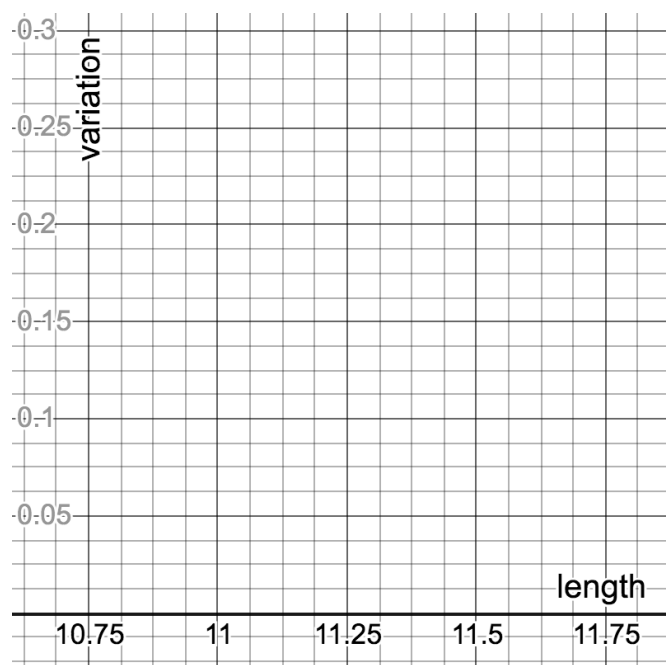
10.2.3 Guided Instruction: Graphing Real-World Applications

1. The average length of a football in the NCAA is $11\frac{5}{32}$ inches, with a tolerance of $\frac{9}{32}$ of an inch (in.). If x is the length of the football, and V is the variation from $11\frac{5}{32}$ in., the function $v(x) = \left|x - 11\frac{5}{32}\right|$ can be used to find the amount of variation.

- a. Determine the minimum and maximum x -values.
- b. Complete the table for $V(x) = \left|x - 11\frac{5}{32}\right|$. Use the minimum x -value in the first row and the maximum x -value in the last row. Add $\frac{3}{32}$ to each x -value starting with the minimum and ending with the maximum.

x	y

- c. Graph the absolute value function.



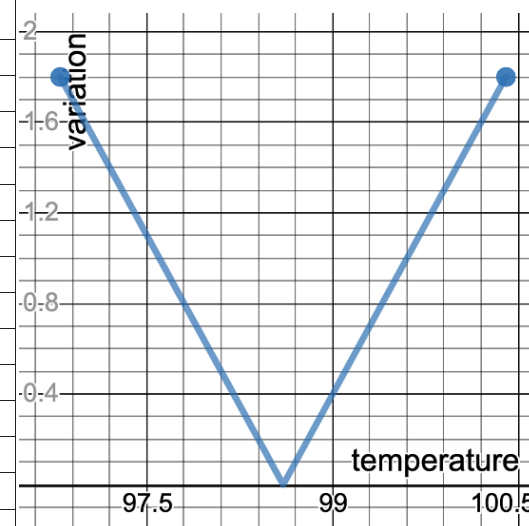
- d. Interpret the vertex of $V(x) = \left|x - 11\frac{5}{32}\right|$.

- e. Interpret the axis of symmetry of $V(x) = \left|x - 11\frac{5}{32}\right|$.

2. For centuries, the average internal body temperature of humans has been designated as 98.6 degrees Fahrenheit. The temperature can vary by as much as 1.8 degrees and still be considered normal. If x is the body temperature, and y is the variation from the average, the equation to find the amount of variation is $y = |x - 98.6|$.

The table of values and the graph for the context is shown.

x	y
96.5	Too much variation
96.8	1.8
97.1	1.5
97.4	1.2
97.7	0.9
98.0	0.6
98.3	0.3
98.6	0
98.9	0.3
99.2	0.6
99.5	0.9
99.8	1.2
100.1	1.5
100.4	1.8
100.7	Too much variation



Complete the table of key features.

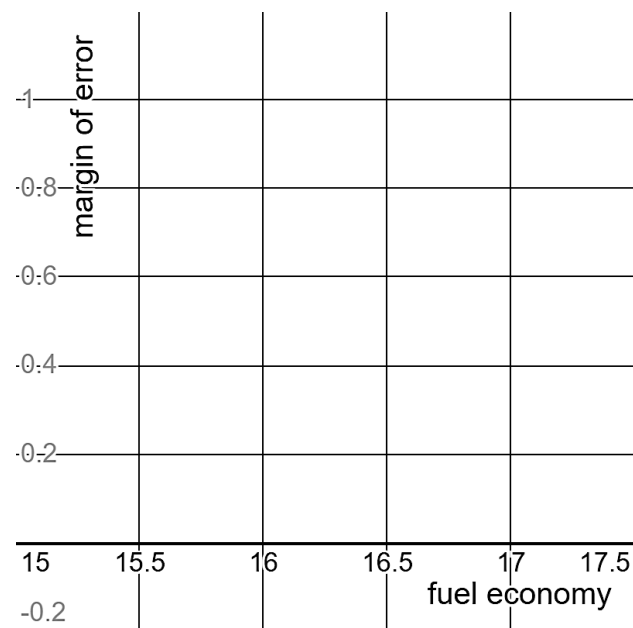
Key Features			
x -intercept		y -intercept	
vertex		line of symmetry	
increasing		decreasing	
domain		range	
opens			
positive		left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{\hspace{1cm}}$
negative		right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{\hspace{1cm}}$



10.2.4 Guided Practice: Graphing Real-World Applications

The 1982 DeLorean DMC 12, the car featured in the movie *Back to the Future*, has an average fuel economy of 16.27 miles per gallon (mpg), according to one study. The margin of error is given as 1.12 mpg. If x is the mileage and E is the amount of error, the function $E(x) = |x - 16.27|$ can be used to find the amount of variation.

- Graph the absolute value function for the expected range of a DeLorean's fuel economy.



2. Complete the table of key features.

Key Features			
x -intercept		y -intercept	
vertex		line of symmetry	
increasing		decreasing	
domain		range	
opens			
positive		left end behavior	$x \rightarrow -\infty,$ $y \rightarrow \underline{\hspace{1cm}}$
negative		right end behavior	$x \rightarrow \infty,$ $y \rightarrow \underline{\hspace{1cm}}$



10.2.5 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

My favorite mistake from today was ...

Means

I

Start

To

Acquire

Knowledge

Experience

Skills